CliniCause: Constructing Reusable Causal-Analysis Datasets from Irregular ICU Records

Anonymous submission

Abstract

1 Introduction

Planned content: problem, resource-level contribution, scope, and bounded contribution summary.

2 Related Work

Planned content: compact gap-led synthesis using only verified bibliography entries.

3 Constructing the CliniCause Datasets

CliniCause constructs two reusable, validated causal-analysis datasets from MIMIC-III and PhysioNet 2012. Each resource connects irregular observations to explicit proxy-state exposures, in-hospital mortality, observed covariates, and inspectable graph metadata. A shared construction workflow provides the same analytical interface while retaining the source-specific meanings of variables, proxy states, and adjustment assumptions. Figure 1 summarizes the construction path, and Table 1 describes the resulting resources.

3.1 Source Cohorts and Analysis Units

The two resources originate from heterogeneous critical-care sources used under their respective access conditions: MIMIC-III, a relational critical-care database, and the PhysioNet/Computing in Cardiology Challenge 2012 collection of ICU time-series records and mortality outcomes (Johnson et al. 2016; Silva et al. 2012). An analysis unit is a source time-series record or ICU stay, not necessarily a unique person. Source record, stay, and admission identifiers are normalized to a canonical `ts_id` join key. Irregular events retain elapsed time, variable identity, and measured value; a record-level table supplies in-hospital mortality and the observed baseline covariates available in that source. Construction preserves two linked views: a long event representation for temporal processing and a record-level representation for causal analysis. Predictive training uses an explicit split-aware form of the former, whereas the causal resource retains the complete admitted identifier collection; validation bridges these forms without treating them as interchangeable artifacts.

The admitted primary causal-analysis populations contain 26,845 MIMIC-III analysis records and 7,993 PhysioNet 2012 analysis records. Their covariate vocabularies, measurement processes, proxy ontologies, and DAGs remain dataset specific. We therefore expose aligned resource interfaces but keep the two datasets and their empirical results separate rather than pooling records or assuming that similarly named constructs are equivalent.

3.2 Proxy-State Design and Deterministic Construction

Proxy-state design begins before patient-level execution. A structured design-time LLM protocol proposed dataset audits, proxy-state ontologies, binary-rule families, missingness considerations, and dataset-specific DAG structures. Project-selected proposals were then encoded in deterministic tagger and graph source. The encoded source, rather than prompt text, is the runtime implementation authority: the LLM neither processes patient records nor assigns exposures or estimates effects during construction.

The resulting dataset-specific rules map available measurements and care-process indicators to canonical binary proxy-state fields. They preserve meaningful source differences—for example, MIMIC-III combines hepatic and coagulation evidence whereas PhysioNet represents them separately. A missing or unavailable numeric measurement contributes no positive rule evidence and is not treated as a normal observation; consequently, input availability remains part of the construct's provenance. These explicit analytical constructs create a transparent interface from irregular records to prediction and downstream analysis, but they are not chart-adjudicated diagnoses.

3.3 Predictive Annotations from Irregular Time Series

For every resource, the predictive task is multi-label estimation of the dataset-specific rule-derived proxy vector. Four learned architectures provide complementary representations: STraTS embeds sparse observation times, values, and variable identities; GRU and TCN process prepared temporal grids with value, observation-mask, and elapsed-time channels; and GRU-D models value and hidden state decay from missingness and time gaps (Tipirneni and Reddy 2022; Cho et al. 2014; Bai, Kolter, and Koltun 2018; Che et al. 2018).

Figure 1 layout placeholder—final paper-native vector figure to replace this box.

Figure 1: CliniCause resource construction. Dataset-specific preprocessing and deterministic proxy rules transform irregular ICU records; four time-series models add normalized predictive annotations, and the aligned proxy layer is packaged with outcomes, observed covariates, and inspectable DAG metadata. Validation gates protect cohort, schema, and provenance integrity before the estimator-ready datasets are used for observational analysis.

Dataset	Records	Exposures	Outcome	Predictive annotations	Causal-analysis estimators
MIMIC-III	26,845	9	In-hospital mortality	1 rule source + 4 models	CF-DML, LinearDML, CausalPFN
PhysioNet 2012	7,993	10	In-hospital mortality	1 rule source + 4 models	CF-DML, LinearDML, CausalPFN

Table 1: CliniCause resource summary for the primary, original-sampling causal-analysis populations. Counts refer to analysis records and admitted non-chronic proxy-state exposures, not raw source-cohort sizes. CF-DML denotes CausalForestDML.

Static covariates are combined with each temporal representation. The targets are proxy states, not mortality.

Each normalized export pairs a probability field with a binary prediction for every proxy target; the implementation threshold is 0.5. Binary-only exports then enter the proxy aggregation interface. In the archived causal runs, this interface combines one deterministic rule-derived source with four aligned model sources—STraTS, GRU, GRU-D, and TCN—through a deterministic majority vote. The aggregate is an algorithmic multi-source proxy-state exposure layer, not a learned label model or expert consensus. Predictive performance characterizes how reproducibly the models transport the rule constructs; it does not establish their clinical validity.

3.4 Causal-Analysis Schema and DAG Metadata

Each estimator-ready analysis row joins the canonical record identifier to the aggregated proxy-state fields, binary in-hospital mortality, and available source-specific covariates. Resource metadata identifies the source dataset and sampling condition and links the table to graph, run, and artifact provenance. The primary analysis admits nine non-chronic MIMIC-III exposures and ten non-chronic PhysioNet exposures; chronic-burden fields remain available as baseline constructs rather than analyzed exposures. The package thus separates the exposure layer from its outcome, adjustment variables, and provenance while keeping their record-level alignment explicit.

For each exposure, a source-coded dataset-specific DAG records project assumptions and maps graph nodes to observed columns to produce an exposure-specific adjustment set. The graphs are assumed structures rather than learned or clinically validated ground truth, but encoding them makes adjustment choices inspectable and replaceable. Researchers can therefore apply alternative estimators, refine proxy definitions, substitute DAGs or adjustment sets, and probe identification assumptions without rebuilding the complete source-data integration pipeline.

3.5 Dataset Validation and Provenance

We define a *validated causal-analysis dataset* through four complementary dimensions. Structural validation checks the expected files, exact column order, canonical identifiers, binary/probability value contracts, and completeness of the output set. Cohort validation requires identifier preservation and exact equality at required handoffs: duplicate or conflicting normalized identifiers, missing or extra records, and silent row shrinkage are rejected rather than resolved by an implicit intersection. These contracts keep one analytical population aligned from deterministic labels through predictive annotations and estimator-ready assembly. Malformed inputs also fail explicitly, including unexpected columns, nonbinary voter fields, out-of-range probabilities, and probability/binary inconsistencies; downstream execution requires the complete expected set of rule and model artifacts.

Artifact and provenance validation binds outputs to metadata, upstream fingerprints or hashes, schemas, cohorts, configurations, and seeds where those records are supported. Run-scoped, dataset-isolated paths prevent cross-source artifact reuse; manifests and stage receipts record planned and completed outputs; checkpoint, prediction, and voter sidecars bind model and input roles; and stale or mutation-mismatched artifacts fail reuse checks. Deterministic transformations and stable derived seeds make repeatable behavior explicit rather than relying on ambient file order or process state.

The evidence layers remain distinguishable. The checked archive establishes schema-valid prediction exports, complete primary aggregate tables, result manifests and checksums, and successful archived analysis-stage records, while its exact producing revisions, configurations, and checkpoint-to-export links are incomplete. The repaired current implementation adds stricter validation contracts supported by source and test cases, but it is not retroactively claimed as the historical producer; current test execution is also not asserted here. This separation allows reuse without erasing known provenance boundaries.

Finally, analytical and empirical validation shows that both resources support the intended observational analysis interface: multi-model predictive characterization, three effect-estimation workflows, DAG-guided adjustment, matching, and robustness diagnostics were applied across the two heterogeneous ICU sources. Validation establishes the structural integrity, cohort consistency, provenance, and analytical reuse of the constructed resources; clinical construct validity and causal identification remain separate scientific questions.

4 Empirical Evaluation

4.1 Prediction Tasks and Metrics

We evaluate a separate multi-label prediction task for each dataset, using its deterministic proxy-state vector as the target and STraTS, GRU, GRU-D, and TCN as the learned models. The comparison uses the selected archived held-out test output for every model–dataset combination. Target sets follow the respective dataset schemas, and neither targets nor scores are pooled across datasets. Because the archive does not establish repeated-run uncertainty, these runs characterize the learnability and reproducibility of the rule-derived constructs rather than statistical superiority or clinical-label validity.

We report binary cross-entropy test loss, macro AUROC, macro area under the precision–recall curve (AUPRC), and macro minRP. For a target, minRP is the maximum across decision thresholds of the smaller of precision and recall. AUROC, AUPRC, and minRP are first computed per proxy target and then averaged; targets containing a single observed class on the evaluated split are excluded from these macro summaries. The loss is the sample-mean multi-label binary cross-entropy recorded by the archived evaluator. Scores are reported as archived point metrics, without unarchived confidence intervals or significance tests.

4.2 Effect Estimators and Cross-Estimator Evaluation

For causal-analysis characterization, each admitted aggregated proxy state is analyzed separately as a binary exposure, with in-hospital mortality as the binary outcome and exposure-specific observed adjustment variables selected through the project DAG. We retain every prespecified non-chronic exposure, without result-based selection, and run the analysis separately for MIMIC-III and PhysioNet 2012. The original, non-downsampled populations define the primary analyses; estimates are not numerically pooled across datasets.

CausalForestDML is the primary nonlinear heterogeneous-effect estimator. It uses double-machine-learning adjustment with flexible nuisance models and a causal-forest final stage to produce record-level fitted conditional-effect estimates (Chernozhukov et al. 2018; Wager and Athey 2018; Athey, Tibshirani, and Wager 2019; Oprescu et al. 2019). LinearDML is the main structured comparator: it retains the DML adjustment framework and nuisance-model family while using a linear final effect model, providing triangulation across final-stage model forms. CausalPFN supplies a distinct complementary modeling perspective and is evaluated over the same prespecified dataset–exposure comparisons to test whether the principal directional patterns are specific to the two DML estimators. It is treated as complementary because the archived pipeline contains less complete estimator-specific sensitivity and permutation support for it than for the DML estimators.

For each dataset–exposure analysis, Results summarize the fitted conditional estimates by their mean over the analyzed sample, termed the *mean model-estimated CATE over the analyzed sample*. This sample mean is not a risk ratio, odds ratio, or unqualified population ATE. Cross-estimator evaluation compares the sign of this quantity for CausalForestDML versus LinearDML and then across all three estimators over all prespecified pairs; disagreement cases are retained explicitly. Directional agreement is a concise robustness summary across model forms, but it neither requires equal magnitudes nor proves causal identification.

4.3 Matching and Robustness Diagnostics

Matching provides a descriptive support-oriented comparison. Binary adjustment variables are retained, other usable numeric adjustment variables are median-binarized, and exposed records are greedily paired one-to-one with available unexposed records by Hamming distance. The matcher begins with exact binary matches and progressively increases the allowed distance to its configured maximum or until its pair-count and match-rate criteria are met. We report the mean within-pair mortality contrast as a *descriptive matched-pair outcome difference*, together with pair availability, final distance, and sufficient-pair warnings. If preprocessing leaves no usable binary matching representation, the comparison fails explicitly rather than being assigned a zero effect.

For the DML analyses, sensitivity evidence retains its archived provenance: estimator-native saved output, later recomputation or labeled reconstruction, partial coverage, and unavailable or failed diagnostics are not treated as interchangeable. Treatment permutation disrupts the analyzed proxy-exposure column, whereas outcome permutation disrupts mortality while preserving the remaining run structure. These are disruption-based sanity checks, not automatically formal randomization tests or identification proofs. Equivalent sensitivity and permutation stages were not archived for CausalPFN, reflecting a difference in the available diagnostic package rather than its role in the cross-estimator comparison.

Outcome-downsampled analyses retain all mortality-positive records and sample mortality-negative records according to the implemented configuration. This changes both the analysis population and mortality prevalence, so it is a robustness test under a major class-balance perturbation rather than a co-primary analysis; its magnitudes are not directly interchangeable with original-population estimates. Applying this complete sequence to both ICU resources evaluates workflow portability under heterogeneous data contracts: the analytical sequence is shared, while definitions and results remain dataset specific and cross-dataset evidence is compared without pooling. Section 5 reports the resulting predictive, estimator-agreement, matching, and robustness findings.

5 Results

5.1 Predictive Characterization

Planned content: archived predictive comparison in a future results table.

5.2 Effect Patterns and Estimator Agreement

Planned content: checked central pattern, named exceptions, and bounded estimator interpretation.

5.3 Robustness and Cross-Dataset Findings

Planned content: checked robustness findings and cross-dataset synthesis.

6 Discussion and Limitations

Planned content: bounded interpretation, explicit failure modes, provenance limits, and release state.

7 Conclusion

Planned content: concise resource-level conclusion with no new claims.

Figure 2 placeholder: cross-estimator directional agreement for 19 dataset–exposure combinations.

Figure 2: Planned Figure 2: cross-estimator directional agreement.

Table 2 placeholder: compact CATE and directional-concordance summary.

Table 2: Planned Table 2: causal-estimation and directional-concordance summary.

References

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